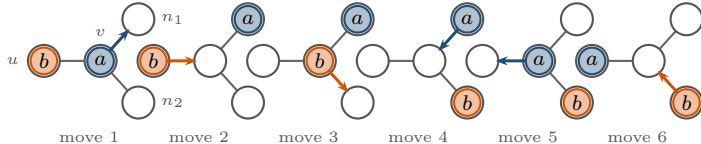
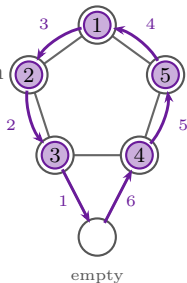


(a) push: a walks to v_4 ; the blockers b and c shift one step down the branch



(b) swap: at a vertex v of degree ≥ 3 with two empty neighbors n_1, n_2 , six moves exchange a and b without moving anyone else



(c) rotate: on a fully occupied cycle, agent 3 steps out into a cleared vertex next to two consecutive cycle vertices, the others advance one position, and agent 3 steps back in behind them: every agent has moved one vertex along the cycle